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Effect of methanolic extract of *Catharanthus roseus* flower in alloxan induced diabetic albino mice through biochemical and metabolic changes assessment.

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Abstract- Diabetes mellitus (DM) is a common endocrine condition that affects how proteins, fats, and carbohydrates are metabolised. It is typified by persistent hyperglycaemia brought on by decreased insulin secretion, action, or both. Traditional treatments for Diabetes mellitus (DM), a serious metabolic condition, involve the use of a number of medicinal plants. Among these is the medicinal herb *Catharanthus roseus*. In the herbal medicine system, the plant has been used to treat diabetes mellitus since ancient times. In preclinical research, the medicinal plant *Catharanthus roseus*, which has long been used to treat diabetes, has demonstrated promise in lowering blood sugar levels and oxidative stress. In this work, mice with diabetes induced by alloxan are used to test the antidiabetic effectiveness of *Catharanthus roseus* flower extract. According to the results, *Catharanthus roseus* flower extract has benefits similar to those of rosiglitazone in lowering blood glucose levels and renal dysfunction indicators such as blood urea, creatinine, SGOT, and SGPT. These results highlight *Catharanthus roseus's* therapeutic advantages and wider applicability in diabetes treatment, hence supporting its potential as a viable botanical therapy for the long-term control of Diabetes mellitus.

Key words: *Catharanthus roseus*, alloxan, albino mice, Diabetes

INTRODUCTION

One of the most prevalent new endocrine disorders is Diabetes mellitus (DM). Due to its high frequency and unsatisfactory clinical treatment results, patients' quality of life decreased. It arises as a result of improper lipid, lipoprotein, and carbohydrate metabolism.[1] According to estimates, this disease would affect over 350 million people globally by 2025, and the potential annual cost of treatment could exceed one trillion dollars. However, by 2035, there will be 109 million diabetic patients in India alone.[2] It will lead to a crisis in public health. According to the study, hyperglycaemia, or an elevated blood glucose concentration, is the condition brought on by the elevated gluconeogenesis activity in the body. The body may experience this as a result of a reduction in insulin secretion brought on by the degeneration of the pancreatic beta cells.[3]

Defects in insulin action, secretion, or both are the cause.[4] The pancreatic beta cells produce insulin, and numerous pathologic mechanisms that contribute to the development of diabetes, such as autoimmune loss of *beta*

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cells, cause blood glucose levels to rise or fall. Numerous organs or organ systems, including the heart, kidney, blood arteries, nerves, and most notably the eyes, are impacted by chronic hyperglycaemia disorders. An increase in the release of insulin from the pancreatic beta cells may be the mechanism causing the blood's insulin activity to rise.[5]

Today, over 25% of prescription medications are derived directly or indirectly from natural sources. Many pharmacologically significant mixtures that are commonly used as home remedies to cure or treat a range of illnesses, from the common cold to malignant development, can be found among South Asia's medicinal plants.[6]

In the ayurveda medical system, *Catharanthus roseus* is a commonly used, readily accessible, and well-recognized medicinal plant that is used as a folk remedy to treat diabetes mellitus. According to reports, this plant's secondary metabolite is used to successfully cure a wide range of illnesses. According to these findings from the literature, *Catharanthus roseus* extract is a dependable natural source of bioactive compounds that, when taken, has beneficial health effects. *Catharanthus roseus* flower extract effectively reduced glucose levels and the oxidative stress brought on by alloxan.

MATERIALS & METHODS

This study includes flower of *Catharanthus roseus* was collected from B. N. College Campus of Patna University, Patna, Bihar, India. It was further identified and botanically authenticated according to the relevant monographs of Indian Pharmacopoeia (2012). For animal studies, the research study was carried out at the Mahavir Cancer Sansthan and Research Centre, Patna, Bihar, India. The albino mice of weighing around 16-20 gram having 6.4 ± 0.5 cm lengths were taken. The mice were housed in polypropylene cages and fed on normal lab made chow and maintained under standard environmental conditions (21±2°C, 55±5% humidity, 12 hr Light: Dark cycle).

To study the induction of diabetes, the Alloxan is purchase from Loba Chemical Pvt. Ltd. was dissolved in 1m Molar citrate buffer at the pH 4.5 and always prepared fresh for immediate use (within 5 minutes). Diabetes was induced by multiple intra-peritoneal injection of freshly prepared alloxan solution in 0.05 M sodium citrate (pH 4.5) at the dose of 35 mg/kg body weight followed by fasting.[7]

To study the animal groupings and experimental design, the mice were divided into five groups having six mice in each group:

Group I - alloxan Induced diabetic control mice receiving citrate buffer only

Group II - non-diabetic control mice receiving only citrate buffer solution

Group III - Diabetic treated (DT100) receiving of 100 mg/ kg of body wt. extract

Group IV - Diabetic treated (DT200) Receiving 200 mg/ kg of body wt. extract

Group V - Diabetic treated (DTRGZ) Receiving 2mg/ kg of body wt. Rosiglitazone

For the extract preparation, freshly harvested leaves samples were washed under running tap water, blotted with filter paper and was dried in the shade at room temperature. The dried plant sample (2.6 kg) was then soaked with absolute methanol under reflux condition for the methanolic extract preparation. The sample was then homogenized with extraction buffer and the supernatant collected after three rounds of extraction. The solvent was evaporated under reduced pressure in a rotary evaporator at 40°C. To this thick paste colloidal silicon dioxide was added and dried in vacuum tube dryer. The obtained plant extract was stored in freezer at -20°C until further test.

For the Biochemical estimation, the desired biochemical parameters were accessed to monitor the metabolic activity of the mice in the respective groups. Fasting Plasma Glucose by GOD/POD method, serum creatinine by alkaline picrate method, Serum urea by Nitro prussic method, Alanine aminotransferase (ALT) Reitman and frankel method and Aspartate aminotransferases (AST) Modified IFCC method.

The statistical analysis, data were expressed as the mean ± S.E.M. for statistical analysis of the data; group means were compared by one-way ANOVA (analysis of variance) with *Post Hoc* analysis. The *Tukey-Karmer Post Hoc* test was applied to identify significance among groups. Graphs are plotted using MATLAB version 7.8.0 R2009a, Natick, Massachusetts: The Mathworks Inc. 2009.

RESULT & DISCUSSION

Acute toxicity and lethal test (LD_{50}) estimation of *Catharanthus roseus*.

The acute toxicity test in albino mice was recorded in table. 1 the acute toxicity testing of *Catharanthus roseus* extract in albino mice gave oral LD_{50} 500 mg/ kg

Table 1: LD_{50} Estimation.

Stage I Treatment	Number used/ number dead
Group 1 (10 mg/kg)	3/0
Group 2 (100 mg/kg)	3/0
Group 3 (1000 mg/kg)	3/1
Stage II Treatment	**Number used/ number dead**
Group 1 (120 mg/kg)	3/0
Group 2 (200 mg/kg)	3/0
Group 3 (390 mg/kg)	3/0
Group 4 (500 mg/kg)	3/1

Table 2: Body weight changes in mice.

Groups	Day 0	Day 7	Day 15
Normal control (NC)	18.85 ± 1.66	21.01 ± 1.29	22.79 ± 1.20
Diabetic control (DC)	12.67 ± 1.08	11.04 ± 0.85	9.47 ± 1.13
Catharanthus roseus flower extract (100 mg/kg) (DT_{100})	12.55 ± 1.98*	12.94 ± 1.36*	13.46 ± 1.32*
Catharanthus roseus flower extract (200 mg/kg) (DT_{200})	11.28 ± 1.66*	13.36 ± 1.84*	14.23 ± 1.64*
Rosiglitazone (2 mg/kg) (DTRGZ)	11.88 ± 2.14*	16.65 ± 1.33*	17.86 ± 2.04*

*Values expressed as Mean ± S.E.M. (Standard Error of the Mean) n=3 in each group: Significant as Compared to Control.

Effect of *Catharanthus roseus* extract on body weight:

Diabetes mellitus has long been treated with the reputedly therapeutic plant *Catharanthus roseus*. It has been demonstrated that consuming *Catharanthus roseus* flower extract improves health by reducing oxidative stress brought on by alloxan, which causes mice's body weight to gradually increase after being lost as a result of diabetes.

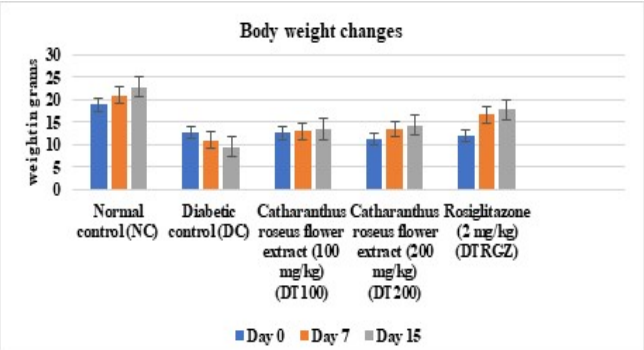


Fig. 1: Effect of treatment with *Catharanthus roseus* flower extract on body weight of different mice groups. NC, untreated normal control animals; DC, diabetic control animals; DT_{100}, diabetic animals treated with a low *Catharanthus roseus* flower extract dose (100 mg / kg b.w.); DT_{200}, diabetic animals' treatment with a high *Catharanthus roseus* flower extract dose (200 mg / kg b.w.). Each bar represents the mean ± SE (n=3).

Table 3: Effects of different doses of *Catharanthus roseus* flower extract and rosiglitazone on blood glucose level in mice.

GROUPS	Blood glucose levels (mmol/L) in week		
	Pretreatment	Post-treatment	
	0 DAY	7 DAY	Day 15
Normal control (NC)	3.85 ± 1.03**	4.06 ± 0.84**	4.03 ± 0.79**
Diabetic control (DC)	14.76 ± 1.45*	14.71 ± 0.85*	14.02 ± 0.68*
Catharanthus roseus extract (100 mg/kg) (DT_{100})	14.34 ± 1.22**	13.87 ± 1.18*	12.74 ± 1.54**
Catharanthus roseus extract (200 mg/kg) (DT_{200})	14.58 ± 1.48**	13.97 ± 1.32**	12.54 ± 1.44**
Rosiglitazone (2 mg/kg) (DT_{RGZ})	16.02 ± 0.85**	10.16 ± 1.37**	9.18 ± 1.36**

*P< 0.05 as Compare with Normal Control. ** P< 0.01 as Compare with Diabetes Control.

Effect of *Catharanthus roseus* flower extract on blood glucose level:

Table 3 shows the variations in blood glucose levels in both normal and diabetic mice before and after therapy. As anticipated, when compared to their normal control counterparts, the DC mice displayed a significantly ($p<0.001$) greater level of glucose. When compared to the DC group, the diabetic mice in both groups (DT_{100} and DT_{200}) displayed lower glucose levels; however, the DT_{200} mice's decrease was more pronounced (-50%; $p<0.001$).

Effect of *Catharanthus roseus* flower extract on blood glucose level

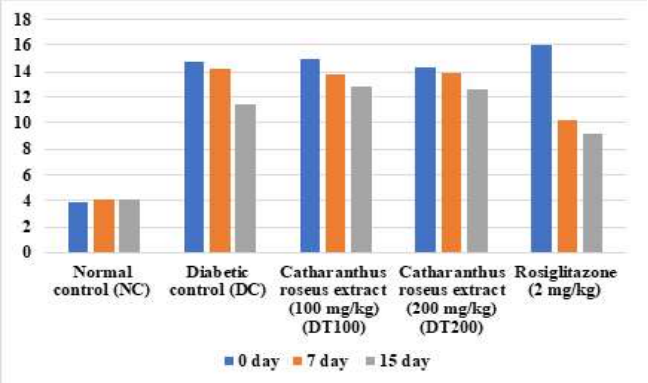


Fig. 2: Effect of treatment with *Catharanthus roseus* flower extract on body glucose levels in different mice groups. NC, untreated normal control animals; DC, diabetic control animals; DT_{100}, diabetic animals treated with a low *Catharanthus roseus* flower extract dose (100 mg / kg b.w.); DT_{200}, diabetic animals' treatment with a high *Catharanthus roseus* extract dose (200 mg / kg b.w.). Each bar represents the mean ± SE (n=3). * P< 0.05 as Compare with Normal Control. ** P< 0.01 as Compare with Diabetes Control.

The DT_{200} group mice's glucose levels were significantly lower than those of the DC group mice during the 4-week treatment program (-68%; p<0.001). However, the DT_{100} rats showed a greater decrease in glucose levels (-57%) than did the DT_{200} mice. On the other hand, following four weeks of treatment, mice in the DTRGZ group displayed an almost 100% decrease in their blood glucose levels (Table 3). These results of the *Catharanthus roseus* flower botanical extract may be attributed to their ability to reduce blood glucose levels by inhibiting the absorption of glucose and improving its storage and utilisation. The results conform to the earlier research.[8,9]

Study of Kidney &Liver markers:

Mice with diabetes have elevated blood urea, creatinine, SGOT, and SGPT levels (about twice as high). When compared to diabetic control mice (DC), all four indicators significantly decline in DTRGZ (rosiglitazone-treated diabetic mice). Serum urea, serum creatinine, serum SGPT, and serum SGOT were all lower in DTRGZ mice. (Table 4) When compared to diabetic control mice, treatment with *Catharanthus roseus* flower extract results in a dose-dependent drop in the levels of all four markers. It was shown that the highest effective dosage was 200 mg/kg of mice's body weight (Table 4). Accordingly, the outcome demonstrated that the flower extract from *Catharanthus roseus* is just as successful in enhancing kidney function as rosiglitazone. *Catharanthus roseus* extract improves histopathological alterations in diabetic rats by lowering inflammation, fibrosis, and apoptosis, hence preserving kidney function towards normal. The earlier findings corroborate the current findings.[10]

Table 4: Blood Urea, Creatinine, SGPT & SGOT in all groups before and after treatment with *Catharanthus roseus* flower extract.

Group	Urea (mg/dl)	Creatinine (mg/dl)	SGPT (IU/L)	SGOT (IU/L)
Normal control (NC)	38.87 ± 0.41**	0.87 ± 0.36**	28.59 ± 0.33**	68.08 ± 1.38**
Diabetic control (DC)	89.08 ± 1.68*	1.32 ± 0.36*	66.68 ± 1.68*	116.30 ± 2.05**
***Catharanthus roseus* extract (100 mg/kg) (DT_{100})**	37.84 ± 1.02**	1.21 ± 0.23 **	44.28 ± 0.54**	78.33 ± 0.22**
***Catharanthus roseus* extract (200 mg/kg) (DT_{200})**	37.76 ± 1.24**	0.92 ± 0.42**	35.62 ± 0.32**	62.2 ± 1.26*
Rosiglitazone (2 mg/kg) (DT_{RGZ})	37.57 ± 0.14	0.96 ± 0.24	29.68 ± 0.54	71.41 ± 1.62

***P< 0.05 as Compare with Normal Control. ** P< 0.01 as Compare with Diabetes Control.**

CONCLUSION

Due to the presence of many active components, which may have distinct modes of action and are reflected in the restored concentration of glucose and organ function enzymes the plant under inquiry, *Catharanthus roseus* flower extract, has anti-diabetic properties. When compared to their normal control counterparts, the DC mice's glucose levels were considerably (p<0.001) higher. When compared to the DC mice, the diabetic mice in both groups (DT_{100} and DT_{200}) displayed lower blood glucose levels; however, the DT_{200} animals' decrease was more pronounced. The results of this investigation showed that *Catharanthus roseus* flower extract use influences the reduction of blood glucose levels. favourably and brings the blood's levels back to normal. This phenomenon may be brought on by biological processes that boost the body's absorption of glucose.

The study demonstrated that an improvement in blood insulin levels may be the cause of the normalisation of blood glucose levels brought on by *Catharanthus roseus* flower extracts administration. The administration of two distinct dosages of *Catharanthus roseus* flower extracts to diabetic albino mice demonstrated a dose-dependent differential protective impact on tests of kidney and liver function. This study demonstrates that *C. roseus* has the potential to be used as a natural oral medication with hypolipidemic and antihyperglycemic properties. Consequently, using botanical phytochemicals as an anti-diabetic treatment may be advantageous and might be regarded as a safe adjunctive treatment for the long-term and successful management of diabetes patients. To clarify the precise mechanism of *C. roseus's* antihyperglycemic activity, further thorough biochemical and pharmacological research is required.

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ETHICAL CLEARANCE STATEMENT

The Current Research Work Was Ethically Approved by the ethical committee of the Mahavir Cancer Sansthan and Research Centre, Patna, Bihar, India.

CONFLICT OF INTEREST

Authors declares no conflict of interest regarding publication or any other activity related to this article.

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